AI-Driven Design of Functional Boron Nitride Materials

Research Plan

Principal investigator, position and affiliated department

Prof. Ma Chen, Assistant Professor, Department of Computer Science, City University of Hong Kong.

Collaborator(s) and respective department(s), if any

Shenzhen Zhongneng Tianyi Technology Co., Ltd. (industrial collaborator).

Duration of the project

Abstract

Boron nitride (BN) materials are promising for wide-band-gap electronics, dielectric layers, thermal management, ultraviolet optoelectronics, and two-dimensional heterostructures. This project aims to discover functional BN and BN-related materials using artificial intelligence. We will construct a BN-centered materials space, learn property models from public materials databases, optimize candidate formulas and structures under multi-objective targets, and prioritize candidates for structure generation and validation. The planned study will cover BN, BCN, BC_2N, SiBN, doped BN, and chemically related analogues. Expected outcomes include a curated BN data layer, AI property models, ranked candidate families, uncertainty-aware selection criteria, first-pass structure hypotheses, and a reusable research platform for academic and industrial collaboration.

Key objectives

1. **Define the BN design space.**

$$\mathcal{C}_{\mathrm{BN}} = \{c : \{\mathrm{B}, \mathrm{N}\} \subseteq \mathrm{elements}(c)\}, \quad c \in \{\mathrm{BN}, \mathrm{BCN}, \mathrm{BC_2N}, \mathrm{SiBN}, \ldots\}.$$

2. **Predict target properties.**

$$f_\theta : (x_{\mathrm{form}}, x_{\mathrm{struc}}) \to \mathbf{y} = (E_{\mathrm{g}}, \Delta H, E_{\mathrm{hull}}, u, \rho_{\mathrm{app}}).$$

3. **Rank and select candidates.**

$$S(c) = w_1 \hat{E}_{\mathrm{g}}(c) - w_2 \hat{E_{\mathrm{hull}}}(c) - w_3 \hat{u}(c) - w_4 d_{\mathrm{dom}}(c) + w_5 \rho_{\mathrm{BN}}(c) + w_6 \rho_{\mathrm{novel}}(c).$$

4. **Generate structure hypotheses.**

$$c \mapsto \{s_{c,1}, \ldots, s_{c,k}\} \mapsto \{\mathrm{geometry}, \mathrm{relaxation}, \mathrm{property\ recheck}\}.$$

Research methodology

1. Candidate universe

The project starts from public 2D materials resources. 2DMatPedia provides an open database of more than 6,000 monolayer structures generated by top-down and bottom-up strategies [1]. JARVIS-Tools supports programmatic access to atomistic materials datasets [2], while Matbench and matminer provide benchmark and featurization foundations for materials ML [3,4]. Additional reference spaces include the Materials Project, C2DB, and related computational materials resources [5,6].

We will represent each record as

$$r_i = (c_i, s_i, \mathbf{y}_i, p_i), \qquad \mathcal{D}_{\mathrm{BN}} = \{r_i : c_i \in \mathcal{C}_{\mathrm{BN}}\},$$

where p_i records provenance. Candidate expansion will combine known BN motifs, same-group substitution, dopant enumeration, and collaborator-proposed chemical families.

2. AI models

The modelling target is not a single score, but a design vector:

$$\max_{c,s} \ \mathbf{g}(c, s) = \big[E_{\mathrm{g}}, \ -E_{\mathrm{hull}}, \ \rho_{\mathrm{dielectric}}, \ \rho_{\mathrm{UV}}, \rho_{\mathrm{2D}}, \ -u\big].$$

Model families will include descriptor models, composition neural networks, crystal graph models, and structure-generation models, drawing on established materials-ML ideas such as CGCNN, ALIGNN, CrabNet, and CDVAE [7–10]. Where suitable, pretrained interatomic-potential models such as CHGNet may support rapid structure screening [11].

3. Candidate selection

The output for each candidate will be a decision vector

$$\mathbf{z}(c) = (\hat{E}_{\mathrm{g}}, \hat{E_{\mathrm{hull}}}, \hat{u}, d_{\mathrm{dom}}, \rho_{\mathrm{BN}}, \rho_{\mathrm{novel}}, a),$$

with action label

$$a \in \{\mathrm{control}, \mathrm{priority}, \mathrm{explore}, \mathrm{hold}\}.$$

This connects model prediction to research action: reference controls, high-priority validation, exploratory families, and candidates reserved for later evidence.

4. Iteration and validation

The project will use grouped evaluation, BN-family evaluation, uncertainty analysis, and rank-stability metrics:

$$\mathrm{MAE},\ \mathrm{RMSE},\ R^2,\ \tau_{\mathrm{Kendall}}, \rho_{\mathrm{Spearman}},\ P(c \in \text{Top-}k).$$

The current local project already provides a starting implementation for public-data ingestion, BN-themed candidate generation, model comparison, rank stability, and first-pass structure handoff. The proposed research will extend this base toward larger BN-relevant datasets, stronger AI models, and structure-level candidate design.

Significance and major deliverables

BN is a high-value family for wide-band-gap and 2D materials research. h-BN is central to ultraviolet optoelectronics and layered-material platforms [12,13], and BN-related analogues create a broad design space for AI-guided discovery. The scientific question is:

Which BN-related compositions and structures should be investigated next?

Major deliverables

1. BN materials dataset: formulas, structures, properties, provenance, and references.
2. AI models for E_{g}, stability proxy, uncertainty, and application-readiness prediction.
3. Candidate families: BN, BCN, BC_2N, SiBN, doped BN, and same-group analogues.
4. Multi-objective ranking: $S(c)$, $u(c)$, $d_{\mathrm{dom}}(c)$, $\rho_{\mathrm{novel}}(c)$, and action labels.
5. First-pass structure hypotheses and validation handoff files.
6. Research outputs: candidate tables, figures, demo, technical report, and manuscript-ready results.

Ethics and safety

The initial project is computational and data-driven. It uses public databases, software, models, and generated candidate structures. If later work involves chemical synthesis or regulated laboratory procedures, the responsible experimental party will handle the required safety approvals.

Selected references

1. Zhou, J. et al. 2DMatPedia: an open computational database of two-dimensional materials from top-down and bottom-up approaches. *Scientific Data* 6, 86 (2019).
2. Choudhary, K. et al. JARVIS-Tools: open-access software for atomistic data-driven materials design. *npj Computational Materials* 6, 173 (2020).
3. Dunn, A. et al. Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm. *npj Computational Materials* 6, 138 (2020).
4. Ward, L. et al. Matminer: An open source toolkit for materials data mining. *Computational Materials Science* 152, 60–69 (2018).
5. Jain, A. et al. Commentary: The Materials Project: a materials genome approach to accelerating materials innovation. *APL Materials* 1, 011002 (2013).
6. Haastrup, S. et al. The Computational 2D Materials Database: high-throughput modeling and discovery of atomically thin crystals. *2D Materials* 5, 042002 (2018).
7. Xie, T. and Grossman, J. C. Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Physical Review Letters* 120, 145301 (2018).
8. Choudhary, K. and DeCost, B. Atomistic line graph neural network for improved materials property predictions. *npj Computational Materials* 7, 185 (2021).
9. Wang, A. Y.-T. et al. CrabNet for explainable deep learning in materials science. *npj Computational Materials* 7, 77 (2021).

10. Xie, T. et al. Crystal diffusion variational autoencoder for periodic material generation. *ICLR* (2022).
11. Deng, B. et al. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence* 5, 1031–1041 (2023).
12. Watanabe, K., Taniguchi, T. and Kanda, H. Direct-bandgap properties and evidence for ultraviolet lasing of hexagonal boron nitride single crystal. *Nature Materials* 3, 404–409 (2004).
13. Cassabois, G., Valvin, P. and Gil, B. Hexagonal boron nitride is an indirect bandgap semiconductor. *Nature Photonics* 10, 262–266 (2016).

Project Details

Project Title: AI-Driven Design of Functional Boron Nitride Materials

PI (Dept): Prof. Ma Chen (Department of Computer Science)

Budget Breakdown

Budget Item	Detail breakdown	Amount (HKD)
Staffing	Student helper / research assistant for data curation, experiments, documentation, and demo maintenance.	
Equipment	Compute, storage, and structure-validation resources.	
General expenses	Cloud credits, backup, software services, hosting, and data-management costs.	
Conference	Presentation at AI-for-science, materials informatics, or applied computing venues.	
Overseas Travel	Optional collaborator visit, conference travel, or partner technical meeting.	
	Total Funding	